

SUMMARY

Electronics and Communication Engineering student with hands-on experience across IoT, Embedded Systems, VLSI, and Computer Vision. Built end-to-end hardware-software solutions using Arduino, ESP32, C/C++, and Python across 4 professional internships and multiple real-world projects. Strong track record in embedded intelligence, automation, and smart systems design.

TECHNICAL SKILLS

- Core Domains** : Internet of Things(IoT), Embedded System, PCB Design, VLSI Design, 3D CAD Design, Computer vision, Image processing
- Languages** : C, C++, Html, Python
- Hardware** : Arduino, Esp32, Sensors, I2C, Webcam
- Tools** : Arduino IDE, Proteus, VS Code, GitHub, AdobePhotoshop, openCV, Blynk, Numpy
- Soft Skills** : Communication & Teamwork, Time Management, Adaptability & Flexibility, Leadership Skills

EDUCATION

2023 - 2027 **B.E. – Electronics and Communication Engineering** - Gnanamani College of Technology | CGPA: 8.0 / 10
2022 - 2023 **H.S. – Maths, Computer Science** - TN State Board | Scored 75.5%

PROJECTS

Smart Home Automation System (IoT) - ESP32, Relay, Wi-Fi Module, Sensors

- Built an IoT-based smart home system for remote monitoring and control of appliances with real-time automation logic.

AI & IoT Based Healthcare Monitoring System - ESP32, IoT Sensors, Python

- Integrated sensors with AI-based analysis to monitor pulse and temperature, enabling real-time health alerts.

Distance Measurement Using Ultrasonic Sensor - Arduino Uno, HC-SR04, LCD Display

- Built a real-time distance measurement system with ± 1 cm accuracy over 2–400 cm, displayed on a 16x2 LCD.

AI-Based Soil Data Communication and Disaster Management - ESP32, Embedded C, Rule-Based AI

- Developed an IoT soil monitoring system with rule-based AI to detect flood/landslide risk and send alerts.

AI Virtual Controller Using Computer Vision - Python, OpenCV, MediaPipe

- Developed a real-time hand-gesture computer controller using Python, OpenCV, and MediaPipe for cursor, click, drag-and-drop, volume, brightness, and screenshot control.

INTERNSHIP EXPERIENCES

Embedded Systems & Computer Vision Intern - NSIC | Chennai

Jul 2026

- Completed hands-on training in Embedded C, sensor interfacing, and circuit implementation; separately trained in Python, OpenCV, image processing, and real-time object detection.

Embedded Systems Intern - UnitedSoftTech | Salem

Dec 2025 – Jan 2026

- Worked on embedded system design using microcontrollers and sensors, implementing real-time embedded programming concepts.

Very Large Scale Integration intern - Smarted Innovation | Bangalore | Online

Oct 2025 – Dec 2026

- Gained practical exposure to VLSI fundamentals, digital logic, CMOS basics, and the IC design flow.

Internet of Things intern - Emglitz Technologies | Salem

Jul 2025

- Applied IoT fundamentals using Arduino and ESP32, integrating sensors for real-time monitoring and control.

Web Development intern - Odugaa Tech | Salem

Jan 2025

- Gained hands-on exposure to front-end fundamentals — HTML, CSS, and basic responsive layout design.

Certifications & Achievements

- PCB Design Workshop** — IndiGuard Systems Pvt. Ltd. at Gnanamani College of Technology. (Sep 2024)
- PROCOMPANION Workshop** — NIT, sponsored by Qualcomm. (Sep 2024)
- National Level Symposium on Generative AI for Electronics and Communication** — KSR Engineering College. (Sep 2024)
- i-SYM'26 (IEEE) Project Expo** — Hindusthan College of Engineering and Technology. (Mar 2026)
- 2nd Prize, Paper Presentation on Haptic Technology** — AVS Engineering College. (Jan 2025)
- Participated in **Smart India Hackathon (SIH)** with an IoT-based solution. (Sep 2025)